HW4

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Pset #4

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Acknowledgements: class notes, Meggie Chen

1:

KNOWN:

- glass: $L_g = 3\text{ mm}$, $k_g = 1.5 \frac{W}{mK}$
- adhesive: $L_a = 0.1\text{ mm}$, $k_a = 1 \frac{W}{mK}$
- silicon: very thin, converts solar energy → elec

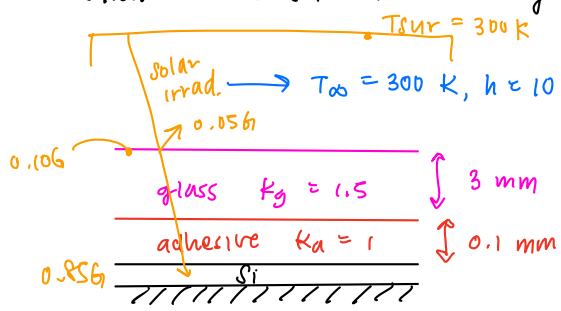
  conversion efficiency: $\eta = a - bT_{si} = 0.553 - 0.001 T_{si}$
- contact thermal res b/w adhesive & silicon: $R_c = 10^{-4} \frac{m^2K}{W}$
- $G = 800 \frac{W}{m^2}$ solar irrad.; 5% solar irrad reflected, 10% solar irrad is absorbed at top of glass, 85% solar irrad transmitted & absorbed w/in Si layer
- $T_\infty = 300\text{ K}$, $h = 10 \frac{W}{m^2K}$
- $T_{sur} = T_\infty = 300\text{ K}$, $h_{rad} = 5 \frac{W}{m^2K}$

FIND:

a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering temp nodes. Calculate value of each thermal resistance in the network

b) Calculate temps of the top surface of the glass $T_{g,top}$ and the silicon T_{si}.

c) Calculate the solar to electricity conversion efficiency η and resulting electricity power

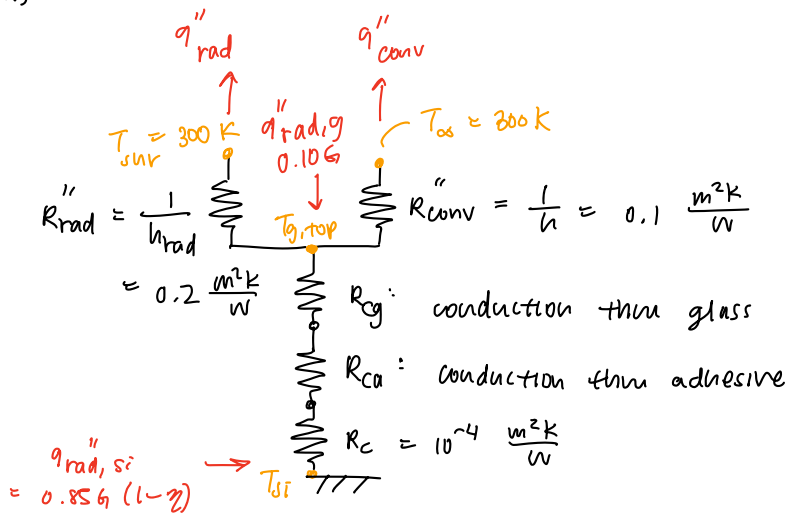


← SCHEMATIC

ASSUMPTIONS: 1D heat transfer; no $\dot{q}$ w/in materials; bottom of Si perfectly insulated; steady state; constant h, k; Si thickness negligible (T doesn't vary across Si)

ANALYSIS:

a)



$R''_{cg} = \frac{L_g}{k_g} = \frac{0.003}{1.5} = 2 \cdot 10^{-3} \frac{m^2K}{W}$

$R''_{ca} = \frac{L_a}{k_a} = 1 \cdot 10^{-4} \frac{m^2K}{W}$

b) $\frac{\Delta T}{R''} = q''$

??

$R''_{cg} + R''_{ca} + R''_c = 0.022 \frac{m^2K}{W} = R''_{cond}$

$R''_{ext} = \frac{1}{\frac{1}{0.2} + \frac{1}{0.1}} = \frac{1}{15} = 0.0667$

energy balance on CV around Si layer:

$$0.85G(1-\eta) = \frac{T_{si} - T_{g,top}}{R''_{cond}} \Rightarrow 680(1-(0.553 - 0.001T_{si})) = \frac{T_{si} - T_{g,top}}{0.0022} \quad (1)$$

energy balance on CV around glass layer:

$$0.85G(1-\eta) + 0.1G = \frac{T_{g,top} - T_\infty}{R''_{conv}} + \frac{T_{g,top} - T_{sur}}{R''_{rad}}$$

$$(2) \quad 0.85(800)(1-(0.553 - 0.001T_{si})) + 0.1(800) = \frac{T_{g,top} - 300}{0.2} + \frac{T_{g,top} - 300}{0.1}$$

solving (1) & (2):

$$\boxed{\begin{aligned} T_{si} &= 342.3 \text{ K} \\ T_g &= 341.1 \text{ K} \end{aligned}}$$

c) $\eta = 0.533 - 0.001T_{si} = 0.553 - 0.001(342.3) = \boxed{0.211}$

electricity power flux: $\eta(0.85G) = 0.190(0.85)(800) = \boxed{143.3 \frac{W}{m^2}}$

2:

KNOWN:

- $D = 10$ mm
- $k = 80 \frac{W}{mK}$
- $T_w = 200^\circ C$
- $L_{ins} = 200$ mm, exposed length L_0 must have temp below $T_{max} = 100^\circ C$
- insulated tip
- convection: $T_\infty = 20^\circ C$, $h = 20 \frac{W}{m^2 K}$

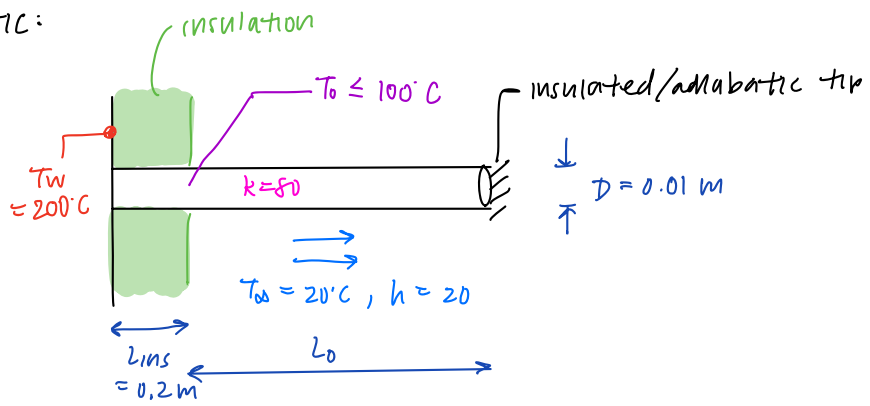
FIND:

a) Derive an expression for exposed surface temp T_0 as a func of the prescribed thermal parameters.

b) If $L_0 = 200$ mm, calculate T_0. Does the exposed length meet specified operating limit?

c) minimum L_0 that ensures T_0 below specified operating limit

SCHEMATIC:



ASSUMPTIONS: L_{ins} & tip perfectly insulated; 1D heat transfer; neglect radiation; steady-state

ANALYSIS:

a) $0 \leq x \leq L_{ins}$: conduction only

$$kA_c \frac{T_w - T_0}{L_{ins}} = q_{cond} \qquad T(0) = 200^\circ C$$

tip condition is adiabatic

$$\hookrightarrow q_f = M \tanh mL \qquad M = \theta_0 \sqrt{hPkA_c} = mkA_c\theta_0$$

$$m = \sqrt{\frac{hP}{kA_c}} = 10$$

$$= mkA_c (T_0 - T_\infty) \tanh mL_0$$

$$\cancel{kA_c} \frac{T_w - T_0}{L_{ins}} = m\cancel{kA_c} (T_0 - T_\infty) \tanh mL_0$$

$$T_w - T_0 = m L_{ins} (T_0 - T_\infty) \tanh m L_0$$

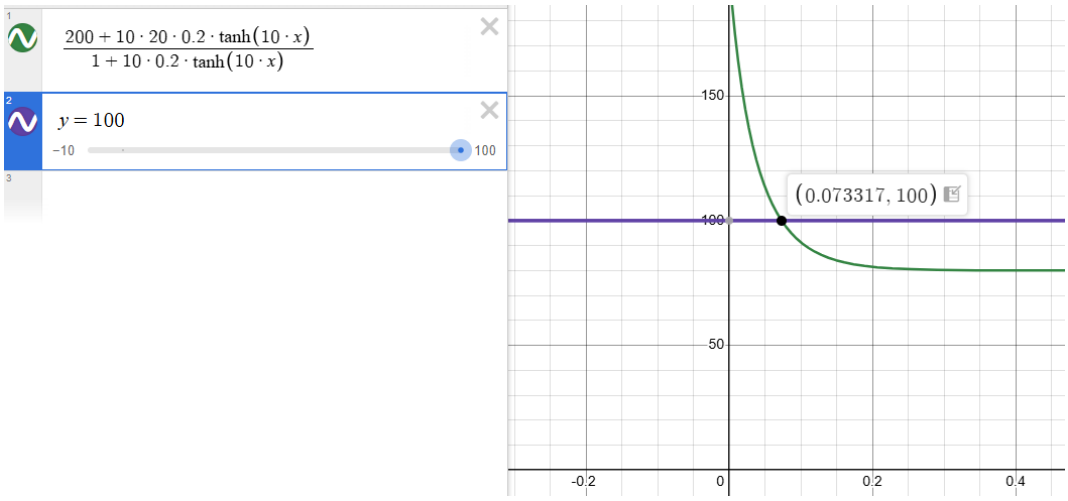
$$T_w + m T_\infty L_{ins} \tanh m L_0 = T_0 (1 + m L_{ins} \tanh m L_0)$$

$$\boxed{T_0 = \frac{T_w + m T_\infty L_{ins} \tanh m L_0}{1 + m L_{ins} \tanh m L_0}}$$

b) $$T_0 (L_0 = 0.2) = \frac{200 + 10(20)(0.2) \tanh(10 \cdot 0.2)}{1 + 10(0.2) \tanh(10 \cdot 0.2)} = \boxed{81.5\,^\circ C \leq 100\,^\circ C}$$

↳ meets operating limit

c) $L_{0,min} = 0.073$ m $= \boxed{73 \text{ mm}}$



3:

KNOWN:

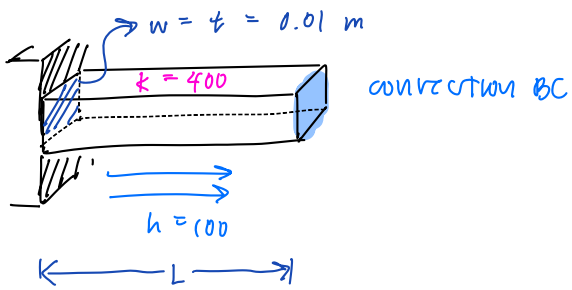
- $k = 400 \frac{W}{mK}$
- exposed to air flow: $h = 100 \frac{W}{m^2K}$; convection tip
- width $s = 10$ mm
- efficiency $\eta_f = 0.6$

FIND:

a) fin length L

b) thermal resistance (R_f), fin effectiveness (ε_f)

SCHEMATIC:



ASSUMPTIONS: 1D heat transfer; steady state; neglect radiation

ANALYSIS:

a) using Table 3.5 (efficiency of common fin shapes):

$$\eta_f = \frac{\tanh mL_c}{mL_c} = 0.6$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{100(0.04)}{400(0.01)^2}} = 10$$

$$L_c = L + \left(\frac{t}{2}\right) = L + 0.005$$

solving for L:

$$0.6 = \frac{\tanh 10(L + 0.005)}{10(L + 0.005)} \Rightarrow \boxed{L = 0.146 \text{ m}}$$

b)

$$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{0.6(100)\underbrace{(2)(0.01)(0.146 + 0.005)}_{A_f = 2wL_c}} = \boxed{5.52 \frac{K}{W}}$$

$$\varepsilon_f = \frac{q_f}{q_{nf}} \qquad \eta_f = \frac{q_f}{q_{max}} \Rightarrow q_f = \eta_f q_{max} = \eta_f h A_f \theta_b$$

$$\varepsilon_f = \frac{\eta_f h A_f \theta_b}{h A_c \theta_b} = \frac{\eta_f A_f}{A_c} = \frac{0.6 \cdot 2(\cancel{0.01})(0.146 + 0.005)}{(0.01)^{\cancel{2}}} = \boxed{18.12}$$